

RAMS: Mechanism-State Residual Operators for Wearable IMTS Forecasting

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Abstract—Wearable IMTS forecasting is difficult because sensor reliability, coverage, and group-asynchronous observation patterns are predictive mechanisms, not incidental missingness. Strong IMTS backbones learn value-level states, but value-similar wearable windows can retain different post-backbone residuals when their mechanism states differ. We propose RAMS, a backbone-agnostic mechanism-state residual operator. Given a trained forecaster F_0 , RAMS freezes F_0 , builds a deployment-valid mechanism state $M = (V, R)$, and adds a closed-gated local/integral residual that is zero at initialization. We formalize mechanism-state aliasing as a residual frontier and instantiate the same operator on tPatchGNN and KAFNet. RAMS-tPatch achieves the best same-protocol reproduced MSE/MAE among public IMTS backbones on MHEALTH, OPPORTUNITY, and PAMAP2, and stays positive under an 8-epoch strong-backbone check. RAMS-KAFNet improves matched frozen KAFNet checkpoints on HumanActivity and the three wearable-suite datasets, winning all 12 paired dataset-seed comparisons on both MSE and MAE. These results show that mechanism-state residuals are a transferable route to more accurate group-asynchronous wearable forecasting. Code is available at this repository: <https://github.com/anonymousUser-dot/RAMS>.

Index Terms—Irregular time series, wearable sensing, human activity, forecasting, temporal patch graphs, residual learning.

I. INTRODUCTION

Irregular multivariate time series (IMTS) are common in mobile health and activity sensing. Due to sensor contact, power management, body placement, and activity-dependent motion, wearable streams often contain asynchronous observations from multiple sensor groups. The goal is not to reconstruct a complete signal for its own sake, but to predict future multivariate sensor values accurately from the observed history.

For this task, a natural strategy is to learn a strong value-level backbone over all observed values. Recent backbones include temporal patch graphs, kernel-aligned forecasters, attention models, and graph models. They are competitive because they summarize irregular histories into predictive latent states. However, such states can still be value-level summaries. When two wearable windows have similar observed values but different coverage, recency, or group reliability, the backbone may map them to similar states even though their forecast residuals differ.

Another natural strategy is to add a generic adapter or residual head after a strong backbone. This can increase capacity, but it does not explain which post-backbone information is allowed to improve prediction. In wearable IMTS, the useful residual information must be available at deployment and tied

to the sensing mechanism: recent reliability, effective counts, time span, coverage integrals, and observed value moments. A generic residual head can therefore learn spurious corrections, while a value-only backbone can miss mechanism-dependent corrections.

Figure 1 gives the roadmap of our solution. Step 1 starts from any trained IMTS backbone F_0 , rather than replacing it. Step 2 extracts deployment-visible mechanism state $M = (V, R)$, where V represents observed value moments and integrals and R represents reliability and coverage. Step 3 constructs local and integral residual features for each forecasted time-channel entry. Step 4 uses a closed-initialized gate to admit a small correction to the backbone forecast. This design is intentionally level-preserving: before residual training, the model is exactly F_0 .

We call the underlying failure mode *mechanism-state aliasing*. Let F_0 be the backbone, $Z = Y - F_0(X)$ its residual, L the value-level state, and M the mechanism state. If $\mathbb{E}[Z | L, M] \neq \mathbb{E}[Z | L]$, then a value-only backbone leaves a residual frontier even when its value dynamics are strong. The squared-loss size of the frontier is

$$\mathcal{F}_M(F_0) = \mathbb{E} \|\mathbb{E}[Z | L, M] - \mathbb{E}[Z | L]\|^2. \quad (1)$$

This identity directly motivates our method: preserve F_0 , expose M to the residual, and keep the residual class small enough that it targets the frontier rather than relearning the backbone.

The main contributions are summarized as follows:

- We define the targeted problem of improving strong frozen-backbone forecasting on group-asynchronous wearable IMTS, and we state the observable conditions used by the model.
- We derive a mechanism-state aliasing theory: the exact squared-loss gap between value-only and mechanism-state residual prediction is $\mathcal{F}_M(F_0)$, and a wearable source condition explains when this frontier is positive.
- We propose RAMS, a level-preserving residual operator that can be attached to a frozen IMTS backbone through protocol-valid mechanism summaries, local/integral branches, and a closed-initialized learned gate.
- We instantiate the same operator on two backbones: RAMS-tPatch for a temporal patch graph and RAMS-KAFNet for a kernel-aligned forecaster. We report shared-protocol public baselines, paired 4-epoch and 8-epoch tPatchGNN results, four-dataset KAFNet paired

results, mechanism-state controls, and theorem-aligned diagnostics.

II. RELATED WORK

Irregular time-series forecasting. Classical IMTS models handle missingness and irregular sampling with recurrent decay, continuous-time latent dynamics, set functions, and attention over observed events [1]–[5]. They are broad, but do not separate nuisance missingness from informative wearable sensing. We follow the missing-data view that an observation mechanism can be informative [6], and specialize it to wearable forecasting.

Graph and patch models. Graph and patch IMTS models, including GraFITi and tPatchGNN [7], [8], represent irregular observations as graph or patch structures. Our baseline scan confirms that tPatchGNN is the strongest reproduced public IMTS backbone under the shared protocol on the main wearable suite. RAMS treats it as one frozen value-dynamics backbone and adds a mechanism-state residual operator.

Missing-value MTS prediction in BigData. LIFE [9] shows that missing-value MTS prediction can benefit from learning features matched to the missingness structure, while joint local/global models [10] emphasize that different temporal scales can matter under missing observations. RAMS addresses a different failure mode: after a strong IMTS backbone has already modeled value dynamics, wearable mechanism states can still split the post-backbone residual inside value-matched windows.

Modern forecasting backbones. General time-series forecasting has advanced through transformer variants, decomposition models, patching, and graph learning [11]–[15]. Recent IMTS work revisits canonical pre-alignment through KAFNet [16], while tPatchGNN avoids sequence explosion by transformable patching and time-adaptive graphs. RAMS is complementary to both: it keeps the chosen backbone fixed and models a deployment-visible mechanism residual that the backbone leaves behind.

Wearable sensing and activity streams. Wearable human-activity sensing has long been studied as a multi-sensor, body-location-dependent recognition problem [17]. The forecasting setting studied here differs from classification, but it inherits the same physical structure: sensor placement, body contact, and activity phase affect which channels are reliable. In activity streams, the same future horizon may be easier or harder depending on whether multiple body-location groups were recently reliable. This motivates treating the wearable observation state as a predictive object rather than as only missing-value nuisance.

Residual forecasting. Adapters and residual heads can correct a trained predictor. Here the residual is tied to a measurable post-backbone error source: mechanism-state aliasing inside value-matched wearable windows. The evaluated target is lower MSE/MAE on a public wearable data family.

III. PRELIMINARIES AND TARGET SETTING

This section defines the data family and prediction problem studied in the paper. We deliberately keep the setting

narrower than general IMTS forecasting: the proposed model is designed for wearable streams whose observation process is informative and whose channels have physical sensor groups.

Definition 1 (Group-asynchronous wearable IMTS). *A group-asynchronous wearable IMTS window is*

$$X = \{(t_j, c_j, m_j, x_j)\}_{j=1}^n, \quad (2)$$

where t_j is time, c_j is a sensor channel, x_j is the observed value, and m_j is a deployment-visible observation indicator. Channels are partitioned into sensor groups, and the observation process may reveal which groups are reliable during the history window. The forecasting target is $Y \in \mathbb{R}^{H \times C}$, evaluated by MSE and MAE.

This family includes the wearable activity datasets used in this paper: MHEALTH [18], OPPORTUNITY [19], PAMAP2 [20], and HumanActivity. They contain multiple body-worn sensor groups and continuous activity streams. In our forecasting conversion, the history window contains asynchronous group observations and the target is the future multivariate sensor trajectory. The main tPatchGNN suite uses the UCI-style group-asynchronous wearable datasets, while the KAFNet instance adds HumanActivity and repeats two main-suite datasets under the same paired operator.

We use three observable conditions to define the target forecasting setting.

C1: grouped sensors. Channels have a physical grouping, such as accelerometer, gyroscope, magnetometer, ECG, or body-location blocks. A channel’s missingness or recency is meaningful only relative to such groups; otherwise reliability features become arbitrary tabular covariates.

C2: mechanism-value non-equivalence. The same value-level trend may occur under different mechanism states. For example, a stable accelerometer segment may arise because the body is stable, or because the relevant device was not recently reliable. A value-only patch state can merge these cases, while the observation measure separates them.

C3: strong patch dynamics. The backbone must already model temporal value dynamics well. If the backbone is weak, improvements from a residual head may simply compensate for missing capacity and will not survive comparison with stronger public models. The method therefore freezes each candidate backbone and measures only paired residual value. In the experiments, tPatchGNN provides the strongest reproduced public backbone on the main wearable suite, while KAFNet provides a structurally different backbone for testing whether the same mechanism operator transfers beyond a temporal patch graph.

These conditions define the evaluated scope: the frozen backbone is already strong, and the remaining residual frontier is coupled to wearable mechanism state. The concrete forecasting conversion and all numerical protocol values are reported in the experimental protocol.

IV. MECHANISM-STATE ALIASING

This section formalizes the failure mode introduced above. A trained IMTS backbone is a value-level predictor when it constructs a state L primarily from observed values, times, and its internal propagation mechanism. In group-asynchronous wearable IMTS, however, coverage, recency, sensor-group reliability, and value moments also carry information about activity phase, device contact, and body pose. The question is whether two windows that are indistinguishable through L can still require different residual corrections once the wearable mechanism state is known.

Let F_0 be a trained backbone and let

$$Z = Y - F_0(X) \quad (3)$$

be its forecasting residual. Let $L = \phi_L(F_0, X)$ denote the value-level state exposed by the backbone, represented in our paired residual interface by the base forecast and forecast-time embedding. Let

$$M = (V, R) = \phi_M(X) \quad (4)$$

be a protocol-valid mechanism state computed only from the observed history. Here V contains observed-history value-distribution information, such as mean, variance, trend, and value integrals, while R contains observation reliability information, such as density, recency, effective count, time span, and coverage integrals. We call M a mechanism state because both parts are computed from the observed history under the wearable observation process: coverage and recency describe the observation process, while value moments describe the observed signal distribution.

Definition 2 (Protocol-valid mechanism state). $M = (V, R)$ is protocol-valid if it uses only quantities available at deployment: observed values, masks, observation times, gaps, recency, coverage, channel identity, and fixed preprocessing normalization constants. It cannot use future targets, validation losses, test losses, or dataset identity as a control input.

Definition 3 (Mechanism-state residual frontier). The post-backbone mechanism-state frontier is

$$\mathcal{V}_M(F_0) = \mathbb{E} \|\mathbb{E}[Z | L, M]\|^2. \quad (5)$$

The part not already captured by L alone is

$$\mathcal{F}_M(F_0) = \mathbb{E} \|\mathbb{E}[Z | L, M] - \mathbb{E}[Z | L]\|^2. \quad (6)$$

$\mathcal{V}_M$ is the ideal squared-loss value of a residual that can use both the base state and mechanism state. It is a population quantity and does not assume that a finite neural module can attain it. $\mathcal{F}_M$ is the stricter aliasing term measured by the diagnostic: it asks whether M explains residual variation among windows with similar value-level state.

Definition 4 (Wearable mechanism-state aliasing). Mechanism-state aliasing occurs when value-level states merge windows that require different residual corrections:

$$\mathbb{E}[Z | L, M] \neq \mathbb{E}[Z | L] \quad (7)$$

on a set of positive probability. In wearable streams this can happen when two windows have similar patch values but different sensor-group coverage, recency, value-moment state, or reliability.

Theorem 1 (Mechanism-state non-identifiability of value-level patch states). Let L be the value-level state exposed by a trained IMTS backbone. Let $\mathcal{A}_L = \{\Delta : \Delta = \Delta(L)\}$ be any value-state-only residual class, and let $\mathcal{A}_{LM} = \{\Delta : \Delta = \Delta(L, M)\}$ be the corresponding mechanism-state residual class with no capacity restriction. Under squared loss,

$$\inf_{\Delta \in \mathcal{A}_L} \mathcal{R}(F_0 + \Delta) - \inf_{\Delta \in \mathcal{A}_{LM}} \mathcal{R}(F_0 + \Delta) = \mathcal{F}_M(F_0). \quad (8)$$

Thus, whenever $\mathcal{F}_M(F_0) > 0$, the value-level patch state is non-identifying for the post-backbone residual: no residual depending only on L can match the risk of a residual that observes the deployment-valid mechanism state M . Moreover, if inside any value-level stratum $L = \ell$ two mechanism states m_a, m_b occur with probabilities p and $1 - p$, and their conditional residual means differ by at least γ , then that stratum contributes at least $p(1 - p)\gamma^2 \Pr(L = \ell)$ to $\mathcal{F}_M(F_0)$.

Proof: The best squared-loss residual using L alone is $\mathbb{E}[Z | L]$; the best residual using (L, M) is $\mathbb{E}[Z | L, M]$. The Pythagorean identity for conditional expectations gives

$$\begin{aligned} \mathbb{E} \|\mathbb{E}[Z | L, M] - \mathbb{E}[Z | L]\|^2 &= \mathbb{E} \|\mathbb{E}[Z | L, M]\|^2 \\ &= \mathbb{E} \|\mathbb{E}[Z | L, M] - \mathbb{E}[Z | L]\|^2, \end{aligned} \quad (9)$$

which is $\mathcal{F}_M(F_0)$. The two-state lower bound follows by conditioning on $L = \ell$: the variance of two conditional means separated by γ with probabilities p and $1 - p$ is at least $p(1 - p)\gamma^2$, and multiplying by $\Pr(L = \ell)$ yields the stated contribution. ■

Theorem 1 is the central residual-frontier identity. The proof uses only the optimality and nesting of conditional expectations under squared loss, so the statement is independent of a particular neural architecture. It states what a value-level backbone state leaves unrecovered in the target family: if two wearable windows have the same patch state but different mechanism-conditioned residual means, then any value-only residual leaves the frontier $\mathcal{F}_M(F_0)$.

Theorem 2 (Group-asynchronous mechanism frontier). Fix a value stratum $L = \ell$. Let $S \in \{0, 1\}$ be a latent wearable mechanism state, such as contact quality or activity phase. Assume that, inside this stratum, mechanism state affects the residual mean through the posterior over S , and suppose

$$\mathbb{E}[Z | L = \ell, S = 1] - \mathbb{E}[Z | L = \ell, S = 0] = \delta_\ell. \quad (10)$$

Let $\eta_\ell(M) = \Pr(S = 1 | L = \ell, M)$. If mechanism state changes the posterior over S inside the stratum, then the contribution of that stratum to the mechanism-state frontier is

$$\Pr(L = \ell) \|\delta_\ell\|^2 \text{Var}(\eta_\ell(M) | L = \ell). \quad (11)$$

Consequently, a positive posterior-variation term and a nonzero residual contrast imply $\mathcal{F}_M(F_0) > 0$.

Proof: Inside $L = \ell$,

$$\mathbb{E}[Z \mid L = \ell, M] = m_{\ell,0} + \eta_{\ell}(M)\delta_{\ell}, \quad (12)$$

where $m_{\ell,0} = \mathbb{E}[Z \mid L = \ell, S = 0]$. The value-only residual mean is the expectation of this quantity over $M \mid L = \ell$. Therefore the within-stratum variance explained by M is $\|\delta_{\ell}\|^2 \text{Var}(\eta_{\ell}(M) \mid L = \ell)$. Multiplying by $\Pr(L = \ell)$ and summing over strata gives the stated contribution to $\mathcal{F}_M(F_0)$. ■

Theorem 2 is the wearable source condition. It specifies why mechanism summaries matter after a strong backbone: $M = (V, R)$ does not need to predict the target directly; it only needs to change the posterior over a mechanism state whose residual contrast remains after value patching. This is the quantity estimated by the split diagnostic in Table X. Without such posterior variation or residual contrast, the mechanism-state frontier vanishes.

Proposition 1 (Finite diagnostic target). *Let $B_L = b_L(L)$ and $B_M = b_M(M)$ be finite value and mechanism bins. Define*

$$D_B = \mathbb{E}\|\mathbb{E}[Z \mid B_L, B_M] - \mathbb{E}[Z \mid B_L]\|^2. \quad (13)$$

Then D_B is the exact squared-loss improvement of the best coarsened-bin residual over the best value-bin residual. If $B_L = L$ and B_M is a coarsening of M , then $0 \leq D_B \leq \mathcal{F}_M(F_0)$. With coarsened B_L , D_B should be read as a coarse frontier proxy rather than a calibrated lower bound.

Proof: The first statement follows from the same conditional-expectation projection identity used in Theorem 1, replacing L, M by B_L, B_M . When $B_L = L$, the residual gain from observing a coarsening B_M cannot exceed the gain from observing the full mechanism state M , giving the upper bound by monotonicity of L_2 projections. If B_L is itself a coarsening, mechanism bins may also absorb variation caused by within-bin value heterogeneity, so the empirical statistic is a proxy for the coarse frontier rather than a formal lower bound on $\mathcal{F}_M(F_0)$. ■

Proposition 2 (Exact residual value under squared loss). *For any residual predictor $\Delta(L, M)$, define*

$$F_{\Delta}(X) = F_0(X) + \Delta(L, M). \quad (14)$$

Under squared loss,

$$\mathcal{R}(F_{\Delta}) - \mathcal{R}(F_0) = \mathbb{E}\|\Delta(L, M) - \Delta^*(L, M)\|^2 - \mathcal{V}_M(F_0), \quad (15)$$

where $\Delta^(L, M) = \mathbb{E}[Z \mid L, M]$.*

Proof: Condition on L, M and expand $\mathbb{E}[\|Z - \Delta\|^2 - \|Z\|^2 \mid L, M]$. The cross term is $-2\langle \Delta, \mathbb{E}[Z \mid L, M] \rangle$, giving $\|\Delta - \Delta^*\|^2 - \|\Delta^*\|^2$. Taking expectation proves the identity. ■

Proposition 2 gives the module-level design rule. The quantity below is not an empirical certificate; it is the population bias-cost decomposition that motivates the finite residual class:

$$\text{improvement} \iff \mathcal{V}_M(F_0) > E_{\text{est}}(\Delta), \quad (16)$$

where $E_{\text{est}}(\Delta) = \mathbb{E}\|\Delta - \Delta^*\|^2$. This is why the model must be small and mechanism-conditioned. A high-capacity residual can make E_{est} exceed the frontier; a weak backbone can leave a large frontier yet still lose to a stronger public model.

Finite-sample implication. The main theory above gives the non-identifiability statement for the target data family. Proposition 2 gives the model-design rule used by the method: a residual should be admitted only when its recoverable mechanism-state frontier is larger than its estimation and optimization cost. In RAMS, that cost is controlled concretely by four restrictions: the F_0 level forecast is frozen, the residual input is the low-dimensional protocol-valid state $M = (V, R)$, both branches are small MLPs over local/integral summaries rather than raw event attention, and the gate is closed-initialized with residual shrinkage. These choices instantiate the finite-sample side of the mechanism-state non-identifiability frontier for any compatible frozen backbone.

Consequences for the architecture. Theorems 1–2 explain the final model choices. First, RAMS starts from a trained backbone F_0 and keeps it fixed, because backbone approximation error should not be solved by a residual. Second, the residual must use protocol-valid mechanism state: Theorem 1 says the value-only gap is exactly the conditional frontier $\mathcal{F}_M(F_0)$. Third, the residual head should be small, regularized, and closed-initialized so that it pays low estimation cost and does not overwrite a strong predictor. Fourth, local and integral branches are both needed because the mechanism-state frontier can come from short-horizon recency, value distribution, or distributed coverage over the whole window.

In short, the theory produces three implementation requirements: preserve the level predictor, encode $M = (V, R)$ rather than raw identity shortcuts, and admit residual capacity only through a small gated operator. Figure 1 shows exactly this operator.

V. RAMS FRAMEWORK

RAMS follows the four-step roadmap introduced in the introduction. First, it obtains a strong value-level forecast from a trained IMTS backbone. Second, it extracts deployment-valid mechanism states from the observed wearable history. Third, it builds local and integral residual features for each forecasted time-channel entry. Finally, it gates the residual correction and adds it to the frozen backbone forecast.

Formally, RAMS takes a trained forecasting backbone F_0 , freezes its value-dynamics forecast, and learns a residual operator whose inputs are restricted to the mechanism state $M = (V, R)$, forecast time, channel identity, and frozen base prediction. For base forecast $\hat{Y}^0 = F_0(X)$,

$$\hat{Y} = \hat{Y}^0 + \Delta_{\text{RAMS}}(X, \hat{Y}^0). \quad (17)$$

The residual is a gated mixture:

$$\Delta_{\text{RAMS}} = (1 - g)\Delta_{\text{loc}} + g\Delta_{\text{int}}. \quad (18)$$

Definition 5 (Mechanism-state residual operator). *For forecast entry (q, c) , let e_q be the forecast-time embedding, $\hat{y}_{q,c}^0$ the*

frozen F_0 prediction, m_c^{loc} the local mechanism summary, and m_c^{int} the integral mechanism summary. RAMS predicts

$$\hat{y}_{q,c} = \hat{y}_{q,c}^0 + (1 - g_{q,c}) a_{q,c}^{\text{loc}} r_{q,c}^{\text{loc}} + g_{q,c} a_{q,c}^{\text{int}} r_{q,c}^{\text{int}}, \quad (19)$$

where

$$\begin{aligned} r_{q,c}^b &= h_b(e_q, \hat{y}_{q,c}^0, m_c^b), \\ a_{q,c}^b &= \sigma(u_b(e_q, \hat{y}_{q,c}^0, m_c^b)), \quad b \in \{\text{loc}, \text{int}\}, \\ g_{q,c} &= \sigma(s(e_q, \hat{y}_{q,c}^0, m_c^{\text{loc}}, m_c^{\text{int}}, r_{q,c}^{\text{loc}}, r_{q,c}^{\text{int}})). \end{aligned}$$

All inputs are deployment-valid functions of the observed history, forecast time, channel, and frozen base forecast. The amplitudes a^b and switch g are initialized near zero, so the initial operator is exactly the backbone and training can only add a learned mechanism-state correction.

The default reported model is the learned gate variant. All ablations disable one ingredient at a time—warmup, open-gate initialization, residual shrinkage, gate target, or boundary regularization—but never choose a different branch per dataset.

Design constraints from the target regime. The method follows four constraints imposed by the target regime: strong IMTS backbones already explain most value dynamics; the residual frontier is small under such a backbone; local and integral branches are both useful but neither dominates uniformly; and linear reliability probes can miss nonlinear residual directions. These constraints lead to the final operator: a frozen backbone plus a compact, softly admitted, mechanism-state residual.

A. Mechanism-State Residual Operator

The method is built around a finite embedding of the wearable observation measure. For channel c , define the deployment-visible empirical measure

$$\mu_c = \sum_{t \in \mathcal{H}} m_{t,c} \delta_{(t/T, x_{t,c})}, \quad (20)$$

where $\mathcal{H}$ is the history window. A value-only backbone mainly uses observed values and times to form L . RAMS adds a separate mechanism-state embedding

$$R_c = [\rho_{\text{loc}}(\mu_c), \rho_{\text{int}}(\mu_c)], \quad (21)$$

where ρ_{loc} contains low-order local moments and recency statistics, while ρ_{int} contains RBF kernel integrals of values and coverage over the whole window. This is the concrete approximation to the mechanism state $M = (V, R)$ in Theorem 1.

The operator is constrained by three design rules. First, it is level-preserving: the output is the frozen F_0 forecast plus a bounded correction. Second, it is mechanism-state only: the residual input is the history observation measure, forecast time, and base prediction, not future loss or dataset identity. Third, it is branch-adaptive: recent local mechanism cues and distributed integral value/coverage cues are trained as separate residual mechanisms before the gate is allowed to select between them. Together these rules turn the residual-frontier identity in Theorem 1 into an implementable forecasting

model: use mechanism state only for the part that value-level patches cannot identify under squared loss.

RAMS operator contract. The operator is not an unrestricted adapter. Its contract is: *inputs* are the frozen forecast $\hat{Y}^0 = F_0(X)$, observed history X , forecast index q , channel c , and deployment-visible summaries of μ_c ; *forbidden inputs* are future targets, test loss, validation selection signals, dataset identity, and any update to F_0 ; *invariant* is $\hat{Y} = \hat{Y}^0$ at initialization; and *allowed action* is only a small additive mechanism-state residual. This contract is shared by RAMS-tPatch and RAMS-KAFNet and is the reason the method can be evaluated by exact paired gains against its own matched backbone.

The resulting operator has a soft-admission structure. Each branch contains its own negative-initialized residual gate, the branch switch is also closed-initialized, and the objective penalizes residual energy. Thus the default prediction is exactly the strong backbone at initialization, and the model must earn every correction through training loss. This matches the strong-backbone setting, where the remaining residual frontier is small.

Soft admission has three nested levels in the implemented model:

$$\hat{Y} = \hat{Y}^0 + (1 - g(q, c)) \alpha_{\text{loc}} \tilde{\Delta}_{\text{loc}} + g(q, c) \alpha_{\text{int}} \tilde{\Delta}_{\text{int}}, \quad (22)$$

with residual-energy regularization in the training objective. The branch amplitudes $\alpha_{\text{loc}}, \alpha_{\text{int}}$ decide whether a correction should be applied; the switch g decides which mechanism scale should dominate; the residual penalty keeps the whole correction close to zero unless it consistently lowers forecasting loss. Here $\tilde{\Delta}_b$ denotes the un-gated branch output and α_b is defined in the branch module below. This nested form is the main methodological difference from a plain additive adapter.

B. Mechanism-State Encoder

For each variable, the local mechanism-state encoder computes the ten statistics used in MechanismSummaryEncoder:

$$\begin{aligned} \rho_{\text{loc}} &= [d, d_{\text{rec}}, \log(1 + \text{count}), \bar{t}, \\ &\quad \sigma_t, t_{\text{rec}}, \bar{x}, \sigma_x, x_{\text{rec}}, x_{\text{rec}} - x_{\text{early}}] \end{aligned} \quad (23)$$

These are exactly the features in the implementation. They capture how much of a channel was observed, whether it was recently observed, how dispersed its observation times are, and whether recent values differ from early values.

The integral encoder appends RBF value and coverage integrals. These are a small kernel-mean embedding of μ_c , chosen because wearable reliability can depend on distributed coverage over the whole history rather than only on the last observation. For RBF center c_m , width w , mask m_t , and value x_t ,

$$\begin{aligned} I_m^x &= \frac{\sum_t \omega_m(t) m_t x_t}{\sum_t \omega_m(t) m_t + \epsilon}, \\ I_m^m &= \frac{\sum_t \omega_m(t) m_t}{\sum_t \omega_m(t) + \epsilon}, \quad \omega_m(t) = e^{-(t-c_m)^2/(2w^2)}. \end{aligned} \quad (24)$$

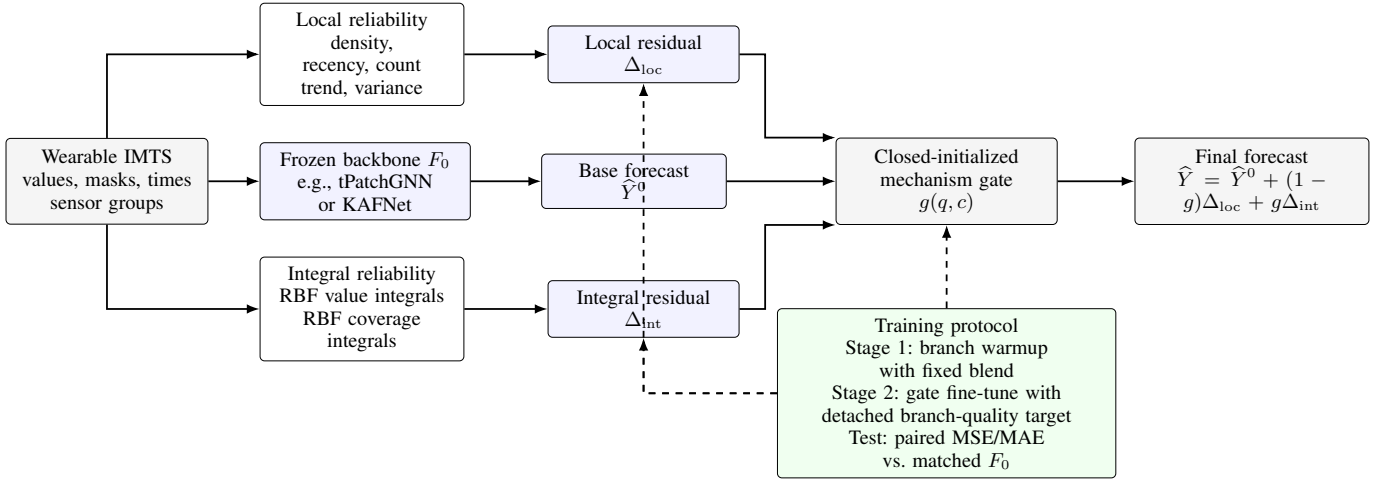


Fig. 1. Overall view of RAMS. A frozen IMTS backbone F_0 provides the level forecast; the paper instantiates this interface with tPatchGNN and KAFNet. Deployment-visible mechanism-state summaries feed local and integral residual branches. A closed-initialized gate selects the residual correction per forecasted time-channel entry. The training protocol is fixed: branch warmup, gate fine-tuning, and paired test MSE/MAE against the matched F_0 checkpoint.

The number of centers, width, and MLP size are fixed by the registered experimental protocol.

C. Local and Integral Branches

Both branches receive a reliability embedding r_c^b , forecast-time embedding e_q , and the base prediction value $\hat{Y}_{q,c}^0$. For branch $b \in \{\text{loc}, \text{int}\}$,

$$\Delta_b(q, c) = \alpha_b(q, c) \cdot h_b([r_c^b, e_q, \hat{Y}_{q,c}^0]). \quad (25)$$

where $\alpha_b(q, c) = \sigma(u_b^\top [r_c^b, e_q])$ is a branch-level admission gate. The local branch uses only ρ_{loc} . The integral branch uses ρ_{loc} plus RBF value/coverage integrals. Both branch gates have negative bias initialization, so their initial residuals are small.

This branch-level gate is the first residual-admission layer: it lets a mechanism feature help where predictive without forcing every time-channel entry to be corrected. The soft-admission ablation shows that the gain is not tied to one gate hyperparameter; RAMS therefore keeps both local and integral mechanism summaries and learns a sample-wise switch above them.

D. Mechanism Gate

The mechanism gate chooses between local and integral correction:

$$g(q, c) = \sigma(s(r_c^{\text{sw}}, e_q, \hat{Y}_{q,c}^0, |\Delta_{\text{loc}}(q, c) - \Delta_{\text{int}}(q, c)|)). \quad (26)$$

The disagreement term is part of the implementation. If the branches agree, the choice has low risk; if they disagree, the gate must use mechanism state to decide whether local recency, value level, or integral coverage is more predictive.

Training has two stages. Stage 1 warms up both branches with a fixed equal blend, independent of the learned switch. Stage 2 learns g with a small boundary penalty $g(1 - g)$ and a detached branch-quality target. The target compares the

training losses of local and integral branch predictions; it uses labels only during training. At deployment, the gate uses only history X , forecast time, and the base forecast.

The gate is the registered default. Its role is stable closed initialization and interpolation between local and integral mechanism corrections. Table VIII reports the effects of removing warmup, boundary regularization, residual L_2 , or the supervised target. The core method is the mechanism-state residual operator; soft admission places it on a strong backbone.

Concretely, let $\ell_{\text{loc}} = (\hat{Y}_{\text{loc}} - Y)^2$ and $\ell_{\text{int}} = (\hat{Y}_{\text{int}} - Y)^2$ during training. The detached gate target in the implementation is a softened branch comparison:

$$T = \sigma\left(\frac{\ell_{\text{loc}} - \ell_{\text{int}} - m}{\tau(\ell_{\text{loc}} + \ell_{\text{int}}) + \epsilon}\right), \quad (27)$$

weighted by confidence $2|T - 0.5|$ with a minimum-confidence threshold. This gives the switch direct supervision about which branch is better on a training example, while preventing test-time label leakage because T is not available or used at deployment.

E. Level-Preserving Residual Operator

The implemented operator is level-preserving: the original F_0 forecast is always present as an additive anchor, and the learned module can only predict a bounded correction. This matters for the target suite. Wearable windows have large value-level dynamics driven by activity intensity. The residual module models the mechanism-dependent residual left after backbone forecasting. We therefore distinguish three paths:

$$\hat{Y}^0 = F_0(X), \quad \hat{Y}_{\text{loc}} = \hat{Y}^0 + \Delta_{\text{loc}}, \quad \hat{Y}_{\text{int}} = \hat{Y}^0 + \Delta_{\text{int}}.$$

Stage-1 warmup trains the two residual paths under a fixed blend so that each branch learns a meaningful correction before the gate can prefer one. Stage-2 gate training then solves a branch-selection problem in residual space. The

Algorithm 1 Registered RAMS training and inference

Require: training windows X, Y , validation split, test split, trained or trainable backbone checkpoint F_0

- 1: Train or load a public IMTS backbone under the shared conversion; keep the matched F_0 checkpoint for each dataset and seed.
- 2: Initialize RAMS with F_0 as the frozen level predictor; freeze all F_0 parameters.
- 3: Build local and integral mechanism-state encoders from deployment-visible history statistics.
- 4: Stage 1: train Δ_{loc} and Δ_{int} with fixed blend $g = 0.5$.
- 5: Stage 2: train learned gate g , residual loss, residual shrinkage, boundary penalty, and detached branch-quality loss.
- 6: Select no branch per dataset: deploy the registered learned gate for all target datasets.
- 7: At inference, compute $\hat{Y}^0 = F_0(X)$, compute Δ_{RAMS} , and output Eq. (17); report paired test MSE/MAE against the same F_0 .

negative branch biases and residual L_2 penalty implement the finite-sample implication of Proposition 2: when the frontier is weak, the learned residual stays close to zero and the output remains close to the backbone.

Train/deploy separation. The branch-quality target T is a supervised training signal, analogous to ordinary labels in a forecasting loss. It is computed on training examples to shape the gate. The deployed gate receives $(R_{\text{loc}}, R_{\text{int}}, e_q, \hat{Y}^0, \Delta_{\text{loc}}, \Delta_{\text{int}})$ and uses no target labels, validation loss, test loss, or realized branch loss.

Backbone-preserving regime. RAMS treats the chosen F_0 as the value-dynamics expert and learns only the mechanism-state residual. The model is evaluated in the regime where strong IMTS backbones already give competitive value forecasts, while wearable mechanism states still split residual behavior inside value-level strata.

F. Training Objective

The training objective is

$$\mathcal{L} = \text{MSE}(\hat{Y}, Y) + \lambda_{\text{res}} \|\Delta_{\text{RAMS}}\|_2^2 + \lambda_{\text{bd}} \mathbb{E}[g(1 - g)] + \lambda_{\text{sup}} \mathcal{L}_{\text{gate}}.$$

All numerical values for the penalties, gate initialization, and residual architecture are listed in the experimental protocol. The ablation disables these ingredients one at a time to test whether the forecasting improvement comes from the mechanism-state residual itself or from a particular admission hyperparameter.

The residual sees protocol-valid mechanism state M , uses local/integral branches, and is admitted through the registered closed-initialized gate. The same default is used for every target dataset.

VI. EXPERIMENTS

A. Experimental Questions

The experiments are designed around the generic $F_0 + \Delta_{\text{RAMS}}$ interface. **RQ1:** Which reproduced public IMTS backbones are strong under the same wearable forecasting conversion? **RQ2:** Does RAMS-tPatch improve a matched frozen tPatchGNN checkpoint on ordinary test MSE/MAE? **RQ3:**

TABLE I

SHARED-PROTOCOL PUBLIC BASELINE SCAN AND COMPACT ABSOLUTE PERFORMANCE ON THE MAIN WEARABLE SUITE. ENTRIES ARE MSE/MAE; LOWER IS BETTER.

Dataset	APN	KAFNet	GraFITi	tPatchGNN	RAMS-tPatch	RAMS-KAFNet
MHEALTH	.761/.484	.751/.478	.736/.464	.705/.446	.696/.438	.718/.452
OPPORTUNITY	.823/.656	.848/.676	.834/.664	.733/.594	.718/.585	.754/.612
PAMAP2	.795/.504	.818/.525	.793/.509	.756/.464	.753/.460	.770/.485

Does RAMS-KAFNet improve a matched frozen KAFNet checkpoint under the same residual protocol? **RQ4:** Which parts of the mechanism-state operator matter: branch capacity, local/integral scale, real alignment, or value/reliability summaries? **RQ5:** Is the mechanism-state aliasing frontier measurable in real wearable data?

B. Datasets, Backbones, and Instances

We evaluate on MHEALTH, OPPORTUNITY, and PAMAP2. These datasets form the target group-asynchronous wearable suite. The public baseline scan includes APNResearch, KAFNet, GraFITi, and tPatchGNN under the same data conversion and metric code. tPatchGNN is strongest on all three target datasets, so the first instantiation, RAMS-tPatch, freezes the matched tPatchGNN checkpoint and trains the RAMS operator on top of it. The forecasting conversion uses history/horizon/patch lengths (250, 50, 25), (300, 60, 30), and (500, 100, 50) for MHEALTH, OPPORTUNITY, and PAMAP2, respectively, with 23, 64, and 37 observed channels. The KAFNet instantiation, RAMS-KAFNet, freezes KAFNet and keeps the same mechanism-state residual operator, optimizer, split rule, metric code, and closed-gate default. The KAFNet paired evaluation covers HumanActivity, MHEALTH, OPPORTUNITY, and PAMAP2, including all three main-suite datasets. This setting tests whether RAMS is a backbone-level residual principle rather than a tPatchGNN-specific module.

For each dataset, seed, and backbone, RAMS is evaluated against the exact matched F_0 checkpoint under the same split, loss, horizon, and metric implementation. Reported gains are paired forecasting gains over that matched frozen checkpoint.

C. Protocol

For every seed, dataset, and backbone instance, we first train or load the matched frozen checkpoint F_0 with the public baseline protocol. For RAMS-tPatch, we report two paired tiers. The 4-epoch tier is the registered scan/ablation protocol because it trains all public baselines and all RAMS branches under one manifest. The 8-epoch tier is the stronger-backbone confirmation: it retraining the tPatchGNN checkpoint longer and repeats the same frozen-backbone gate procedure. For RAMS-KAFNet, we report the available three-seed frozen-KAFNet paired rows on HumanActivity, MHEALTH, OPPORTUNITY, and PAMAP2. In all cases RAMS is trained for three residual epochs with MSE loss and frozen F_0 . Batch size follows the registered manifest; for the KAFNet confirmation it is 16 on HumanActivity and 32 on MHEALTH, OPPORTUNITY, and PAMAP2. The UCI wearable runs use `collate_fn_patch` and dataset-specific patch lengths: 25 for

TABLE II
REGISTERED FORECASTING PROTOCOL. THESE SETTINGS MATCH THE
EXPERIMENT SCRIPT AND ARE KEPT FIXED ACROSS THE TARGET SUITE.

Item	Setting
Backbone interface	frozen F_0
Backbone update	frozen
Public baselines	APNResearch/KAFNet/GraFITi/tPatchGNN
Backbone instances	RAMS-tPatch on tPatchGNN; RAMS-KAFNet on KAFNet
Residual module	shared RAMS mechanism-state operator
Default variant	learned gate
Residual epochs	3
Base epochs	4 scan/ablation; 8 strong confirmation
Loss	MSE
Batch size	manifest value; KAFNet 16/32
Optimizer / LR	Adam / 10^{-3}
Split	81% train, 9% validation, 10% test
Patch length	25/30/50 for MHEALTH/OPP/PAMAP2
History/horizon	250/50, 300/60, 500/100
RBF centers / width	6 / 0.18
Residual penalties	10^{-4} residual, 2×10^{-4} boundary
Gate initialization	branch -3.0 , switch -2.5
Gate supervision	weight 0.02, fixed 0.5 warmup blend
Seeds	three paired seeds per reported dataset
Pairing	exact matched F_0 checkpoint per dataset-seed
Metrics	test MSE, test MAE

MHEALTH, 30 for OPPORTUNITY, and 50 for PAMAP2. The model is evaluated on the same test split as its own base checkpoint. No test metric is used to train the gate.

The processed wearable records use a fixed record-order split: 81% train, 9% validation, and 10% test, with no shuffling. Windows are generated with the repository activity-chunk stride of 4000 raw time units. Continuous channels are standardized once by feature-wise observed-value mean and standard deviation before applying the group-asynchronous mask; every baseline and RAMS run consumes the same processed tensors. All wearable runs use Adam, learning rate 10^{-3} , delayed-step decay, validation interval 1, patience 2 in the registered RAMS runs, and one training iteration per seed.

Both tiers use the same residual architecture. The mechanism-state encoder computes the ten local statistics in Sec. V, appends six RBF value/coverage integrals with width 0.18, and passes the local and integral vectors through two-layer GELU MLPs with layer normalization. Stage 1 uses a fixed 0.5 blend for branch warmup. Stage 2 uses residual penalty $\lambda_{\text{res}} = 10^{-4}$, branch gate initialization -3.0 , switch initialization -2.5 , boundary penalty $\lambda_{\text{bd}} = 2 \times 10^{-4}$, and supervised gate weight 0.02.

The default model is fixed before aggregation: the closed-initialized gate. The ablations test which parts of the admission procedure affect the result. The target metric is ordinary test MSE/MAE.

Aggregation rule. All averages are computed over the registered dataset-seed list in the experiment manifest. A seed contributes a paired row only if both the matched F_0 checkpoint and the RAMS checkpoint finish under the same split, horizon, and metric implementation. The gate is fixed before reading aggregate results, while no-warmup, open-gate, no-residual- L_2 , no-gate-target, and no-boundary rows are soft-admission ablations. Public baselines are trained under the same conversion before any RAMS aggregate is interpreted.

Implementation and overhead. RAMS adds no graph construction, raw-event attention, or backbone update. Its overhead is a mechanism-state scan, six RBF value/coverage inte-

grals, and two small MLP residual branches. For C channels, history length T , horizon H , $M = 6$ RBF centers, and residual width d , the additional work is $O(CT + CTM + CHd^2)$, while the backbone graph or attention computation is unchanged.

D. Backbone-Parallel Paired Forecasting Results: RQ1–RQ3

Table I answers RQ1 and separates absolute performance from backbone transfer. tPatchGNN is the strongest reproduced public IMTS backbone under the shared conversion, RAMS-tPatch gives the lowest main-suite MSE/MAE, and RAMS-KAFNet shows the same operator improves a different frozen backbone. Table III answers RQ2 for the tPatchGNN instance. A single RAMS gate improves the matched tPatchGNN checkpoint on every dataset-metric average. Across the nine dataset-seed pairs, RAMS-tPatch wins eight times on MSE and eight times on MAE. Because the comparison is paired against the exact matched checkpoint modified by RAMS, the gains measure residual forecasting value rather than differences in splits, horizons, or preprocessing.

Table IV reports paired uncertainty and mean gains. The registered three-seed protocol is paired at the checkpoint level: each RAMS-tPatch model is compared with the exact tPatchGNN checkpoint it modifies. The evidence has two stable positives and one smaller-margin positive. MHEALTH and PAMAP2 are positive for both metrics in every seed, with positive bootstrap intervals within the three-seed rows. OPPORTUNITY improves in mean, with MSE gains ranging from -0.79% to 1.50% and MAE gains from -0.03% to 0.49% . At the suite level, the nine paired rows give positive aggregate evidence for both MSE and MAE.

TABLE V
RAMS-tPatch 8-EPOCH STRONG-BACKBONE CONFIRMATION. ENTRIES
SHOW PAIRED BASE-TO-RAMS VALUES, GAIN PERCENTAGES, AND SEED
WINS.

Dataset	MSE base→ours	Gain	Wins	MAE base→ours	Gain/Wins
MHEALTH	0.701129→0.693718	1.048%	3/3	0.443561→0.435146	1.877% / 3/3
OPPORTUNITY	0.708311→0.703887	0.620%	2/3	0.575071→0.572392	0.466% / 3/3
PAMAP2	0.756257→0.752854	0.449%	3/3	0.467224→0.460352	1.460% / 3/3

Table V reports the 8-epoch strong-backbone confirmation for RQ2. The stronger 8-epoch tPatchGNN lowers the raw backbone error on all three datasets, yet the same frozen-backbone gate remains positive on all six average comparisons, with 8/9 MSE and 9/9 MAE seed wins. The effect sizes shrink as the base predictor absorbs more of the residual frontier, while the direction remains consistent.

E. KAFNet Backbone Results: RQ3

Table VI answers RQ3. HumanActivity, MHEALTH, OPPORTUNITY, and PAMAP2 are all three-seed KAFNet paired evaluations, and RAMS-KAFNet wins all paired seeds on both MSE and MAE over the matched frozen KAFNet base. The main-suite KAFNet rows are especially strong: MHEALTH improves by 4.40%/5.82%, OPPORTUNITY by 11.73%/9.99%, and PAMAP2 by 5.63%/6.97% in MSE/MAE. Table VII adds the aggregate paired uncertainty: all twelve rows are positive, with bootstrap intervals strictly above zero. Together, Tables III and VI show the intended use of RAMS:

TABLE III

RAMS-tPatch REGISTERED 4-EPOCH PAIRED SCAN RESULTS. RAMS USES ONE FIXED DEFAULT: THE LEARNED GATE. IT IMPROVES THE STRONGEST REPRODUCED PUBLIC IMTS BACKBONE UNDER THE SHARED PROTOCOL (tPatchGNN) ON ALL AVERAGE MSE/MAE COMPARISONS IN THE MAIN WEARABLE SUITE.

Dataset	n	tPatchGNN MSE	RAMS-tPatch MSE	MSE gain	MSE wins	tPatchGNN MAE	RAMS-tPatch MAE	MAE gain	MAE wins
MHEALTH	3	0.705	0.696	1.19%	3/3	0.446	0.438	1.81%	3/3
OPPORTUNITY	3	0.733	0.718	2.11%	3/3	0.594	0.585	1.56%	3/3
PAMAP2	3	0.756	0.753	0.33%	3/3	0.464	0.460	0.78%	3/3

TABLE IV

PAIRED AGGREGATE EVIDENCE FOR THE 4-EPOCH REGISTERED PROTOCOL OVER THE NINE DATASET-SEED PAIRS. BOOTSTRAP p IS ONE-SIDED $\Pr(\bar{g} \leq 0)$ FROM PAIRED RESAMPLING; SIGN p IS THE EXACT ONE-SIDED SIGN TEST.

Metric	Mean gain	95% boot CI	Wins	$p_{\text{boot}}/p_{\text{sign}}$
MSE	1.210%	[0.330, 2.110]	9/9	< 0.001/0.002
MAE	1.383%	[0.780, 1.810]	9/9	< 0.001/0.002

TABLE VI

RAMS-KAFNet PAIRED RESULTS. THE FROZEN BACKBONE IS KAFNet, WHILE THE SAME MECHANISM-STATE RESIDUAL GATE, LOSS, SPLIT, AND METRIC CODE ARE USED.

Dataset	n	KAFNet MSE/MAE	RAMS-KAFNet MSE/MAE	MSE gain	MAE gain
HumanActivity	3	0.044 / 0.121	0.042 / 0.115	4.08%	5.14%
MHEALTH	3	0.763 / 0.491	0.718 / 0.452	5.85%	7.89%
OPPORTUNITY	3	0.848 / 0.678	0.754 / 0.612	11.10%	9.66%
PAMAP2	3	0.816 / 0.525	0.770 / 0.485	5.63%	7.62%

the same mechanism-state residual operator can be attached to different trained IMTS backbones, and the reported gain is always measured against the exact frozen checkpoint it modifies.

Dataset-level results. For RAMS-tPatch, MHEALTH and PAMAP2 are the stable positives: both win all three seeds on both metrics in the 4-epoch protocol, and both remain positive after the backbone is strengthened. OPPORTUNITY has positive mean gains with smaller seed-level margins and one negative MSE edge case in the scan protocol. The diagnostic in Table X reports the residual-variance share used to quantify the observed mechanism-state frontier in each dataset.

F. Soft-Admission and Mechanism Controls: RQ4

Table VIII separates the branch choices from the input state controls. Local-only and integral-only residuals are already positive, confirming that both short-range mechanism summaries and distributed coverage/value integrals carry usable residual information. The registered gate has the best suite-average MSE/MAE gain, so we keep it as the fixed deployment rule rather than selecting branches per dataset.

TABLE VIII

BRANCH/GATE ABLATION AGGREGATED OVER THE NINE REGISTERED tPatchGNN DATASET-SEED PAIRS. ENTRIES ARE PAIRED GAIN PERCENTAGES AND SEED WINS OVER THE SAME FROZEN CHECKPOINTS.

Variant	MSE gain	MSE wins	MAE gain	MAE wins
Local	0.804	9/9	1.343	8/9
Integral	0.843	9/9	1.397	8/9
Fixed mix	0.835	9/9	1.400	8/9
Gate	0.851	8/9	1.482	8/9
Conservative gate	0.798	8/9	1.471	8/9

TABLE VII

RAMS-KAFNet PAIRED AGGREGATE EVIDENCE OVER ALL TWELVE COMPLETED DATASET-SEED ROWS. BOOTSTRAP p IS ONE-SIDED PAIRED RESAMPLING; SIGN p IS THE EXACT ONE-SIDED SIGN TEST.

Metric	Mean gain	95% boot CI	Wins	$p_{\text{boot}}/p_{\text{sign}}$
MSE	6.665%	[4.080, 11.100]	12/12	< $10^{-5}/2.44 \times 10^{-4}$
MAE	7.578%	[5.140, 9.660]	12/12	< $10^{-5}/2.44 \times 10^{-4}$

We further run input controls that keep the same frozen tPatchGNN checkpoint, training schedule, and gate, while replacing the residual input state. The *value-only* control removes mask density, recency, effective counts, time span, and coverage integrals; the *reliability-only* control removes value moments and value integrals; the *shuffled* control batch-rolls the mechanism state so that marginal scale remains but sample alignment is broken. Table IX shows two facts. First, value summaries, reliability summaries, and residual capacity each contribute to the residual operator. Second, the full aligned state is best or essentially tied on MHEALTH and PAMAP2, while OPPORTUNITY has smaller margins. The shuffled control preserves residual-input scale but breaks the sample-specific M -to-residual link in Theorem 1.

TABLE IX

MECHANISM-STATE CONTROLS. ENTRIES ARE PAIRED MSE/MAE GAIN PERCENTAGES OVER THE SAME FROZEN tPatchGNN CHECKPOINT.

Dataset	Real	Value	Reliab.	Shuffled
MHEALTH	1.346/2.082	1.202/1.976	0.976/1.833	0.991/1.870
OPPORTUNITY	0.524/ 0.214	0.715 /0.161	0.600/-0.055	0.596/-0.060
PAMAP2	0.683 /2.152	0.654/ 2.165	0.435/1.422	0.425/1.387

G. Mechanism-State Aliasing Diagnostic: RQ5

We also estimate whether the theory’s mechanism-state frontier exists before training the neural residual. For each wearable window, we compute a deployment-visible value-level summary L and mechanism summary $M = (V, R)$. We then discretize L into value-level strata and ask whether mechanism bins split the future level residual inside the same value stratum. This is the empirical counterpart of Theorem 1 and Proposition 1: if mechanism state does not change the conditional residual after value matching, the coarse frontier statistic should be near zero.

The diagnostic is an ANOVA-style coarse frontier statistic

$$D_{\text{split}} = \sum_{\ell} \hat{p}(\ell) \sum_r \hat{p}(r | \ell) \|\hat{\mu}_{\ell,r} - \hat{\mu}_{\ell}\|_2^2, \quad (28)$$

where $\hat{\mu}_{\ell,r}$ is the mean future residual in value stratum ℓ and mechanism bin r . We report both D_{split} and its percentage of residual variance. Because the value state is binned rather

than observed exactly, this is a coarse proxy for $\mathcal{F}_M(F_0)$. It is computed after model evaluation and checks whether mechanism state splits value-matched residuals in the target suite.

TABLE X
MECHANISM-STATE ALIASING DIAGNOSTIC. AFTER VALUE MATCHING,
MECHANISM BINS EXPLAIN NONZERO RESIDUAL VARIATION.

Dataset	n	D_{split}	Residual-var. share	Mean gap
MHEALTH	2021	5.34×10^{-3}	0.789%	0.060
OPPORTUNITY	785	8.17×10^{-3}	1.522%	0.087
PAMAP2	902	6.23×10^{-3}	1.024%	0.069

The diagnostic answers RQ4 directly. All three datasets have a nonzero value-stratified mechanism split, accounting for roughly 0.79–1.52% of residual variance. These are small numbers, which is consistent with the fact that tPatchGNN is already a strong patch-graph forecaster; nevertheless they are of the same order as the paired MSE gains in Table III. This closes the theory-to-experiment loop: deployment-visible mechanism state splits value-matched residuals, and a compact nonlinear mechanism residual recovers a portion of that frontier in forecasting MSE/MAE.

H. Reproducibility and Diagnostics

The registered 4-epoch per-seed signs match the dataset-level reading: MHEALTH and PAMAP2 are positive in every paired seed on both metrics, while OPPORTUNITY has one negative MSE seed and one near-zero negative MAE seed. The diagnostic in Table X uses five value-summary quantile bins, three mechanism-summary quantile bins, and a minimum cell size of eight windows. All reported tables are constructed from paired CSV summaries produced by the same forecasting pipeline: the same dataset, seed, split, horizon, metric implementation, and matched frozen F_0 checkpoint are used for every gain. The baseline scan is aggregated before any RAMS rows; Tables III, IV, V, and VI use the corresponding paired rows; and Tables VIII–X change only the residual variant, residual input, or diagnostic readout.

VII. CONCLUSION

In this paper, we proposed RAMS, a mechanism-state residual operator for forecasting group-asynchronous wearable IMTS. The key idea is to preserve a trained IMTS backbone as the value-level predictor and construct a small residual correction from deployment-visible mechanism state. The mechanism-state aliasing identity explains when such a correction can reduce forecasting risk: if value-matched wearable windows have different mechanism-conditioned residual means, a value-only patch state leaves a recoverable residual frontier.

RAMS implements this idea with local reliability summaries, integral value/coverage summaries, and a closed-initialized gate. Instantiated on two frozen backbones, RAMS-tPatch improves matched tPatchGNN checkpoints on MHEALTH, OPPORTUNITY, and PAMAP2 in both 4-epoch and 8-epoch protocols, while RAMS-KAFNet improves matched KAFNet checkpoints on HumanActivity,

MHEALTH, OPPORTUNITY, and PAMAP2, winning all 12 reported paired dataset–seed rows on both metrics. Ablations, mechanism-state controls, and aliasing diagnostics show that the gains come from aligned wearable mechanism information rather than generic residual capacity.

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